

The background is a solid dark blue color, overlaid with white line art illustrations of various cosmetic products and people. At the top, there are brushes in a holder, bottles, a hand holding a face mask, and a woman applying makeup. On the left, there's a large bottle of spray and a jar. On the right, a woman is shown applying lipstick. At the bottom, there are more brushes, a woman applying makeup to another woman's face, and a large, stylized eye graphic.

DPEDAL FINAL PROJECT PRESENTATION

BRIDGING THE SKIN TONE INCLUSIVITY GAP IN THE INDIAN COSMETICS INDUSTRY

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Introduction

The Indian cosmetics industry has rapidly expanded, yet a significant gap remains in shade inclusivity. Most international and domestic brands often cater disproportionately to lighter skin tones, leaving individuals with medium to deeper complexions underserved. This lack of inclusivity not only limits consumer choice but also reinforces colorism in beauty standards.

Moreover, AI-powered beauty apps and recommendation engines frequently misclassify Indian skin tones due to biased datasets trained primarily on Western complexion ranges. This results in inaccurate product suggestions and poor consumer trust. Therefore, there is a critical need for a data-driven approach that evaluates shade ranges across brands.

Problem *statement*

The Indian cosmetics industry has rapidly expanded, yet a significant gap remains in shade inclusivity. Most international and domestic brands often cater disproportionately to lighter skin tones, leaving individuals with medium to deeper complexions underserved. This lack of inclusivity not only limits consumer choice but also reinforces colorism in beauty standards.

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Problem *objective*

- **Market size:** India's beauty and personal care market is projected to cross \$30+ billion by 2030, with foundation sales growing rapidly in e-commerce.
- **Consumer frustration:** A survey by Nykaa revealed that over 65% of Indian women struggle to find their correct foundation match
- **Inclusivity gap:** Despite India's diversity, many brands still launch limited shades (10–15) compared to 40+ shades abroad.
- **Cultural impact:** Fairness creams have historically dominated the market → inclusivity in makeup is not just cosmetic, but also a social statement.

We chose this problem because it lies at the intersection of technology, inclusivity, and real consumer need.

Literature *review*

S.NO.	NAME	YEAR	AUTHOR	FOCUS OF THE PAPER
1	Skin Tone Estimation under Diverse Lighting Conditions	2024	S.K. Mbatha	They build a CNN-based skin tone estimation model trained on 21,375 images across varied pigmentation. They show regression models can achieve good accuracy across lighting conditions.
2	Evaluation of the Human Visual System in Cosmetics: Foundation Colour Selection	2020	Altynay Kadyrova, Majid Ansari	This is a psychophysical experiment evaluating how humans match skin colour to foundation shades under different illuminants. They show that people tend to pick lighter/paler shades and that instrument-based matching is far better

Literature *review*

S.NO.	NAME	YEAR	AUTHOR	FOCUS OF THE PAPER
3	Foundation Makeup Shade Recommendation using Computer Vision Based on Skin Tone Recognition	2023	Sanica Kamble	This work uses skin detection to extract RGB values from face photos across different ethnic groups. They compute the difference between calculated & actual skin-tone.
4	An Investigation of Colorism and Gender Imbalance in AI in the Indian Context	2024	E. Vijit	Exploratory analysis of how Generative AI models (Text-to-Image) amplify existing stereotypes of colorism and sexism when generating images of Indian people based on various prompts.

Methology

1. DATA COLLECTION

- **Brand Shade Data:** The system uses CSV files from multiple cosmetic brands such as L'Oréal, MAC, and Maybelline, each containing pixel-level color data representing various foundation shades.
- **User Skin Data:** Captured in real time using a webcam. Facial landmarks are detected to extract consistent skin regions (cheeks, forehead, nose bridge, chin).

2. SKIN TONE DETECTION

- **The script `face_color_palette.py` uses:**
 - Dlib's 68-point face landmark predictor to identify regions of interest.
 - OpenCV to capture real-time frames and extract RGB pixel data from selected face regions.
 - KMeans clustering (from scikit-learn) to determine the top 5 dominant skin colors from those regions.\
- **The detected color palette (user's skin tones) is:**
 - Displayed as a color strip.
 - Saved as both an image (`saved_palette.png`) and a JSON file (`saved_palette.json`).

Methology

3. PRODUCT PALETTE EXTRACTION

- Implemented in `closest_palette_match.py`.
- Each foundation shade image data (from CSV) is parsed using regex and converted into a structured RGB array.
- KMeans clustering identifies dominant tones for every product.
- These palettes are compiled brand-wise and saved into a single JSON file, `processed_brand_palettes.json`.

4. COLOR MATCHING ALGORITHM

`model.py` loads both:

- The user's skin tone palette from `saved_palette.json`.
- The processed brand palettes from `processed_brand_palettes.json`.

5. COLOR MATCHING TECHNIQUES

- Color Space Conversion
- Distance Computation
- Best Match Selection

Dataset: *Brands' websites*

Official brand websites give us a reliable foundation shade reference to train the model, validate predictions, and offer cross-brand recommendations that consumers can actually buy.

M.A.C

UK

🔗 <https://www.maccosmetics.co.uk/product/13847/12O613/products/makeup/face/foundation/studio-fix-fluid-spf-15-foundation?shade=NC1O>

INDIA

🔗 <https://www.maccosmetics.in/product/studio-fix-fluid-spf-15-24hr-matte-foundation-oil-control?shade=NW6O>

MAYBELLINE

USA

<https://www.maybelline.com/face-makeup/foundation-makeup/fit-me-matte-poreless-foundation?variant=334>

INDIA

🔗 <https://www.maybelline.co.in/all-products/face/foundation/fit-me-foundation?variant=Natural-Buff>

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L'ORÉAL

USA

[https://www.lorealparisusa.com/makeup/face
/foundation-makeup/true-match-super-
blendable-makeup](https://www.lorealparisusa.com/makeup/face/foundation-makeup/true-match-super-blendable-makeup)

INDIA

[https://www.lorealparis.co.in/true-match/super-
blendable-liquid-foundation-2r2c2k-rose-
vanilla](https://www.lorealparis.co.in/true-match/super-blendable-liquid-foundation-2r2c2k-rose-vanilla)

Tools & libraries used

CATEGORY	TOOLS/LIBRARIES
PROGRAMMING LANGUAGE	PYTHON
IMAGE PROCESSING	OPENCV, DLIB
MACHINE LEARNING	SCIKIT-LEARN (KMEANS)
NUMERICAL COMPUTATION	NUMPY
COLOR SCIENCE	SCIKIT-IMAGE (RGB2LAB), SCIPY
DATA HANDLING	PANDAS, JSON
VISUALIZATION	OPENCV WINDOWS, COLOR PALETTE IMAGES
OPTIMIZATION	HUNGARIAN ALGORITHM (LINEAR SUM ASSIGNMENT)

Techniques

- **Facial Landmark Detection:** Used to locate consistent skin sample regions.
- **KMeans Clustering:** Extracts dominant RGB tones from facial pixels and product shades.
- **Color Space Transformation (RGB → LAB):** Improves matching accuracy.
- **Optimal Assignment:** Ensures minimal total color distance between palettes.
- **Automated Workflow Execution:** Managed through a single script (Found_ai_tion.py).

Results *& discussion*

- » The system successfully extracts five dominant tones from a user's face.
- » Foundation shades from multiple brands are represented as color palettes.
- » The LAB-based distance metric provides perceptually realistic matches.
- » Tests showed that the LAB Colour matching significantly outperformed naive RGB matching in terms of perceived color accuracy.
- » Visual comparison using OpenCV confirms that matched foundations closely resemble the user's detected tone range.
- » Limitations include:
 - Variations in lighting conditions during webcam capture.
 - Dependence on pre-existing brand shade data.
 - Facial landmark model dependency (shape_predictor_68_face_landmarks.dat).

Conclusion

This project presents a computer-vision-based skin tone detection and foundation recommendation system. By combining facial landmark analysis, KMeans clustering, and LAB color matching, it accurately identifies the most suitable foundation shades from various brands. The system promotes inclusivity and personalization in the cosmetics industry, bridging the gap for underrepresented skin tones and ensuring better product satisfaction.

References

- Dlib Library Documentation – <http://dlib.net>
- OpenCV Documentation – <https://docs.opencv.org>
- Scikit-learn Documentation – <https://scikit-learn.org>
- Scikit-image Documentation – <https://scikit-image.org>
- SciPy Linear Sum Assignment (Hungarian Algorithm) – https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html
- Pandas Developer Guide – <https://pandas.pydata.org>

THANK YOU!